

# Minimal building blocks for introspection in self-referential theories

An answer to the minimality question after Corollary 23 of S. A. Alexander, *Self-referential theories* (2020)

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**Correction (27 September 2026).** The stratified analogue of Lemma 19 is Lemma 65 of *Self-referential theories*, not Lemma 64, which concerns validity. The minimality result is unaffected. As stated, it holds among the three named schemas, under the source’s own operator semantics.

**Status.** Hand proofs with explicit countermodels. Not machine-checked; no independent review. The literature was not checked for priority.

**Source.** S. A. Alexander, *Self-referential theories*, Journal of Symbolic Logic 85(4), 2020. [arXiv:2008.11535](https://arxiv.org/abs/2008.11535), [doi:10.1017/jsl.2020.54](https://doi.org/10.1017/jsl.2020.54).

**The question.** After Corollary 23: “It would be interesting to investigate questions about whether the above building-blocks are minimal. For example, in Lemma 19, is it really necessary to bundle j-Introspection with all three other schemas? For now, we will leave those questions open.”

**Answer.**

- **No: i-Validity is redundant.**  $[AV]_i \cup [i\text{-Deduction}]_i \cup [i\text{-Introspection}]_j$  is already closed-r.e.-generic.
- **The other two are both needed.** Dropping Assigned Validity (AV) or i-Deduction destroys genericity. So  $\{AV, i\text{-Deduction}\}$  in theory  $i$  is the **unique minimal** bundle that makes i-Introspection in theory  $j$  generic, among subsets of Lemma 19’s three schemas.

## Setting (the paper’s §§2-3)

- **The base logic.** The operator  $T_i$  is interpreted in a structure by an arbitrary truth value  $M \models T_i\varphi[s]$ , subject only to Definition 2’s three conditions: independence of non-free variables, alphabetic variants, and substitution of *variables*.
- **Validity** means truth in all such structures.
- **The substitution instance**  $\varphi^s$  replaces free variables by numerals, including inside operators, so  $(T_i\varphi)^s \equiv T_i(\varphi^s)$ .
- **The intended structure**  $M_U$  of a family  $U$  sets  $M_U \models T_i\varphi[s] \iff U_i \models \varphi^s$ .
- **Closed-r.e.-generic** means:  $M_U \models T$  for every closed (each  $U_i$  contains  $T_i\sigma$  for  $\sigma \in U_i$ ), r.e. family  $U \supseteq T$ .
- **The schemas:**
  - AV =  $\{\varphi^s : \varphi \text{ valid}\}$ ;
  - i-Validity (iV) =  $\text{ucl}(T_i\varphi)$  for  $\varphi$  valid;
  - i-Deduction (iD) =  $\text{ucl}(T_i(\varphi \rightarrow \psi) \rightarrow T_i\varphi \rightarrow T_i\psi)$ ;
  - i-Introspection (iI) =  $\text{ucl}(T_i\varphi \rightarrow T_iT_i\varphi)$ .

**The fact that makes AV non-trivial.** Validity is *not* closed under substituting numerals inside operators, because Definition 2 links  $T_k\phi(x|y)$  with  $T_k\phi[s(x|s(y))]$  only for variables  $y$ .

- *Example.*  $\chi(x) \equiv \forall y (y = x \rightarrow (T_k\theta(y) \leftrightarrow T_k\theta(x)))$  is valid, by condition 3.
- But  $\chi(n) \equiv \forall y (y = n \rightarrow (T_k\theta(y) \leftrightarrow T_k\theta(n)))$  is not valid: nothing ties the atom  $T_k\theta(n)$ , with its numeral, to the value of  $T_k\theta(y)$  at  $y = n$ .

## What introspection needs

For an assignment  $s$ ,  $M_U \models (T_i\phi \rightarrow T_iT_i\phi)[s] \iff (U_i \models \phi^s \Rightarrow U_i \models T_i(\phi^s))$ . Every sentence is some  $\phi^s$ . So:

$M_U \models \mathbf{i\text{-}Introspection} \iff U_i \models \psi$  implies  $U_i \models T_i\psi$ , for every sentence  $\psi$ .

## Theorem 1. $[AV]_i \cup [iD]_i \cup [iI]_j$ is closed-r.e.-generic

It is r.e. by the paper's Theorem 3. Let  $U \supseteq$  the family be closed and r.e.

- **The AV and iD parts** are true in  $M_U$ , by the paper's Lemmas 18(1) and 17.
- **For iI.** Suppose  $U_i \models \psi$ . By compactness (Theorem 3(4)),  $\chi \equiv \sigma_1 \rightarrow \dots \rightarrow \sigma_n \rightarrow \psi$  is valid for some  $\sigma_k \in U_i$ .
  - *The key step.*  $\chi$  is a valid sentence, so  $\chi = \chi^s \in AV \subseteq U_i$ . Closure then gives  $T_i\chi \in U_i$ , and also  $T_i\sigma_k \in U_i$  for each  $k$ .
  - Repeated i-Deduction then gives  $U_i \models T_i\psi$ . ■

The paper's proof of Lemma 19 obtains  $T_i\chi$  from i-Validity. AV already puts the valid *sentence*  $\chi$  into  $U_i$ , and closure supplies  $T_i\chi$ . So i-Validity is used only to make i-Validity itself true in  $M_U$ . That is why Lemma 19 needs it in its own bundle, but introspection does not need it.

## Proposition 2. Without i-Deduction, genericity fails

The family is  $F = [AV]_i \cup [iV]_i \cup [iI]_j$  (or anything smaller). Take the least closed family  $U \supseteq F$ , so  $U_i$  is the  $T_i$ -closure of  $AV \cup iV$ , plus  $iI$  if  $i = j$ .

**The countermodel.** Let  $X$  be the closure of  $U_i \cup \{\chi^s : \chi \text{ valid}\}$  under  $\sigma \mapsto T_i\sigma$ , taken up to alphabetic variants. Let  $M$  have standard first-order part, interpret  $T_i\phi[s]$  as " $\phi^s \in X$ ", and interpret every other operator similarly by its own theory.

This is a legitimate base-logic structure. Its truth value depends only on  $\phi^s$ , which gives conditions 1 and 2. And  $(\phi(x|y))^s = \phi^{s(x|s(y))}$ , which gives condition 3. Check that  $M \models U_i$ :

- *Substitution.*  $M$  interprets formulas by substitution, so  $M \models AV$  (as in Lemma 8 / 18(1)).
- *i-Validity.*  $M \models \text{ucl}(T_i\chi)$  for  $\chi$  valid, since  $\chi^s \in X$ .
- *Closure members.*  $M \models T_i\sigma$  for  $\sigma \in U_i$ , since  $\sigma \in X$ , and likewise for iterates.
- *Introspection.* The  $iI$  instances hold, since  $X$  is closed under  $T_i$ .

**The failure.** Let  $\psi \equiv T_i(0=0) \wedge T_i(1=1)$ .

- $U_i \models \psi$ , since both conjuncts are  $iV$  instances.
- But  $\psi \notin X$ : it is not in  $U_i$ , not an instance of a valid formula (its atoms are free), and not of the form  $T_i\sigma$ . So  $M \not\models T_i\psi$ , hence  $U_i \not\models T_i\psi$ .
- By the criterion above,  $M_U \not\models \mathbf{i\text{-}Introspection}$ . ■

## Proposition 3. Without AV, genericity fails

- **If iV is kept** ( $[iV]_i \cup [iD]_i \cup [iI]_j$ ): the iV part is already false in  $M_U$ . Take  $k \neq i$  and the  $\chi$  above. Then  $\text{ucl}(T_i\chi(x)) \in U_i$ , but  $M_U \models T_i\chi[x \mapsto n]$  needs  $U_i \models \chi(n)$ . Here is a model of  $U_i$  falsifying  $\chi(n)$ :
  - make  $T_i\phi[s]$  true for every  $\phi$  and  $s$ , which satisfies iV, iD, iI and all closure members;
  - make  $T_k\phi[s]$  true unless  $\phi$  is literally  $\theta(n)$ .
 This respects Definition 2, whose condition 3 concerns variables only. In it  $T_k\theta(y)[y \mapsto n]$  is true while  $T_k\theta(n)$  is false, so  $\chi(n)$  fails. Hence  $U_i \not\models \chi(n)$ .
- **If iV is also dropped** ( $[iD]_i \cup [iI]_j$ , or  $[iI]_j$  alone): take the least closed  $U$ .
  - $U_i \models 0=0$ , trivially.
  - Let  $D$  be the closure of  $U_i$  under modus ponens and under  $\sigma \mapsto T_i\sigma$ .  $U_i$  contains only iD instances,  $T_i$ -prefixed formulas, and iI instances if  $i = j$ . So every member of  $D$  is a  $T_i$ -formula or an implication between such, and **(0=0)  $\notin D$** .
  - The model  $T_i\phi[s] : \iff \phi^s \in D$  satisfies:
    - \* the iD instances, because  $D$  is closed under modus ponens;
    - \* the closure members;
    - \* the iI instances, because  $D$  is closed under  $T_i$ .
 But it makes  $T_i(0=0)$  false, so  $U_i \not\models T_i(0=0)$ .
  - Introspection fails at  $\phi = (0=0)$ . ■

## Corollary

Among the subsets of  $\{AV, iV, iD\}$  placed in theory  $i$ , those making  $[\cdot]_i \cup [iI]_j$  closed-r.e.-generic are exactly those containing **AV and iD**:  $\{AV, iD\}$  and  $\{AV, iV, iD\}$ . The unique minimal one is  $\{AV, iD\}$ .

## Scope, and the second question

- The analysis covers Lemma 19's schemas only. With other schemas in the pool (Peano arithmetic, SMT, the  $T_j\phi$  ( $\phi \in T_j$ ) schema) there could be other minimal bundles. **Conjecture G1**: none of Corollary 23's other blocks can replace AV or iD for introspection.
- The stratified analogue (Lemma 65 / Corollary 71; corrected from "Lemma 64") should admit the same reduction: drop the validity schema, keep assigned validity and deduction. It was not checked here.
- The paper's second question, how far the stratified generic construction can be strengthened, is open-ended and untouched.